State Banks as Shock Absorbers?

Relationship Lending, the Cost of Debt, and Investment during India's 2026 Energy Shock*

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Abstract

Do relationships with state-owned banks cushion firms when an external shock compresses margins? We study India's 2025–26 energy shock — the October 2025 U.S. sanctions on Rosneft and Lukoil, which eliminated the discount on Russian crude, followed by the February–June 2026 Strait of Hormuz crisis, which drove Brent from \$62 to a monthly average of \$117 and the rupee to a record low — as a sharp, externally imposed, and unevenly felt shock to Indian corporate costs. Extending the government-bank-lending framework of Lin et al. (2017) from crisis-era Japan to contemporary India, we compare listed firms whose pre-shock banking relationships are predominantly with public sector banks (PSBs) to observably similar firms banking privately, in a difference-in-differences and triple-difference design with firm and industry×time fixed effects, where the third difference is each firm's pre-shock energy intensity. Relative to the Japanese evidence, we add the *price* margin of credit — the effective cost of debt — to the quantity and investment margins, and we test whether any state-bank cushion flows to viable constrained firms or to firms with

*We are grateful for the framework of Lin et al. (2017), which this paper explicitly extends, and to seminar participants for early comments. An interactive companion illustrating the identification design is available at `https://1boscochiramel.github.io/state-banks-in-a-storm/`. All errors are our own. **Working paper: a pre-specified research design with a public-data pilot study.** The pilot results in Section 6 are estimated entirely from public disclosures and inference is reported both cluster-robust and by exact permutation test; the full-study tables remain pre-specified skeletons pending access to the CMIE Prowess universe and are explicitly marked as such.

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interest coverage below one. This draft develops the institutional setting, data infrastructure, and identification strategy in full, and reports a *pilot study* on 74 listed firms built entirely from public disclosures — exchange filings for financials, and, for the treatment, annual-report banker lists together with the *amount-weighted* bank-lender annexures of CRISIL credit rating rationales. In the pilot, the effective cost of debt of state-bank-attached firms rises about 0.6–1.1 percentage points less than that of private-bank firms after the shock, a cushion appearing only post-shock. Because seventeen treated firms make cluster-robust asymptotics unreliable, inference is by *exact Fisher randomization test*: the amount-weighted cushion has permutation $p = 0.047$ (binary: $p = 0.062$), the capex cushion $p = 0.035$, and placebo shock dates one year earlier show nothing. The estimate is negative across every treated-group size we built (seven to seventeen firms) — the sign is robust — though magnitude varies with sample composition, and we report the full, most-inclusive sample rather than a more favourable intermediate one. A pre-committed equity event study finds no cushion in raw returns but a significant one *among the most energy-exposed firms* at the sanctions announcement. Estimation on the full CMIE Prowess universe is presented as pre-specified table skeletons.

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1 Introduction

In fiscal year 2025, advances by India's public sector banks (PSBs) grew 12.2 percent against 9.5 percent for private banks — the first fiscal year state banks out-lent their private rivals since FY2010.[1] Within months, Indian industry was hit by the sharpest external cost shock in a generation: United States sanctions on Rosneft and Lukoil (October 2025) erased the discount at which India had been importing Russian crude, and the closure of the Strait of Hormuz (February–June 2026) drove Brent crude from roughly \$62 to a monthly average of \$117 per barrel and the rupee to record lows beyond Rs. 95 per dollar. The coincidence of a state-led credit expansion with an abrupt, uneven squeeze on corporate margins reproduces, with striking fidelity, the configuration that Lin et al. (2017) exploit in Japan's early-1990s crisis — and it raises the same question in a new economy and a new decade: *when private credit conditions tighten, do firms attached to state-owned banks fare differently, and if so, is that a correction of a market failure or a misallocation of capital?*

Economic theory offers two well-known and opposing answers. In the *social* view, government banks exist precisely for moments like this: insulated from the funding and capital pressures that force private lenders to retrench, they can lend counter-cyclically into a credit crunch and relax constraints for viable firms (Hoshi et al., 1991; Khwaja and Mian, 2008). In the *political* view, state banks are instruments of patronage whose lending responds to elections and connections rather than opportunity (La Porta et al., 2002; Sapienza, 2004; Dinç, 2005; Cole, 2009); in a crisis, their marginal rupee flows to the connected and the moribund, and the aggregate consequence is the "zombie" dynamic documented for Japan by Peek and Rosengren (2005) and Caballero et al. (2008). The two views agree that state banks will lend *more* in bad times; they disagree about *who* receives the credit and *what it produces*. Crises are therefore the natural arena in which the theories can be told apart.

The sharpest existing crisis test is Lin et al. (2017). Using a registry-quality panel of loans outstanding from each individual bank to each listed Japanese firm, they show that a one-yen increase in lending by Japan's nine fully government-owned banks was associated with ¥0.84 of additional firm investment in normal times and a further ¥0.51 during the 1990–94 crisis; that the effects concentrated in credit-constrained firms — smaller, more levered, outside the *keiretsu* main-bank system, and in industries dependent on external finance (Rajan and Zingales, 1998); and that, contrary to the zombie-lending presumption,

[1] Growth rates computed from Reserve Bank of India data (SBI Research compilation); see Section 2.2.

firms with interest coverage below one were *less* likely to receive government credit during the crisis. Government bank lending in Japan, on their evidence, had a "bright side."

This paper asks whether the bright side travels. We port the design to India's 2025–26 energy shock and extend it in three directions. First, we add the *price* margin of credit. The Japanese evidence concerns quantities and investment; we ask whether state-bank attachment shows up in what firms *pay* — the effective cost of debt, measured as interest expense over average borrowings — as well as in what they spend on capital formation. Second, we move the crisis test to India, where the normal-times result is already established: Srinivasan and Thampy (2017) show that relationships with government-owned banks relax investment–cash-flow sensitivities for Indian firms in tranquil periods. Whether that relationship value survives, grows, or reverses under stress is precisely the cell of the $\{\text{Japan, India}\}\times\{\text{calm, crisis}\}$ matrix that remains empty. Third, the shock we study is contemporary, which brings both a cost and an opportunity: the cost is that our post-period is short and partially reversed — the June 17, 2026 Islamabad accord reopened Hormuz and Brent retraced to roughly \$74 — which is why our design is quarterly; the opportunity is that the estimates speak to a live policy debate rather than a settled historical one.

Identification follows the logic of the Japanese study, adapted to Indian data realities. India has no public loan registry, so our treatment is the *identity of a firm's bankers*: CMIE Prowess lists the banks with which each firm maintains relationships, and since the 2020 consolidation India has exactly twelve public sector banks, making the classification of each relationship unambiguous. A firm is PSB-attached if the majority of its bankers, measured *before* October 2025, are state-owned; freezing the treatment pre-shock ensures that crisis-driven switching cannot contaminate it. Outcomes are the cost of debt and the capital-expenditure ratio. The baseline design is a difference-in-differences around the arrival of the shock with firm and industry×quarter fixed effects; the central specification is a triple difference in which the third margin is each firm's pre-shock energy intensity (power and fuel expense over sales). A confounder capable of mimicking our triple-difference coefficient would need to affect PSB-attached firms, only after October 2025, and in proportion to their pre-shock energy dependence — a demanding conjunction that excludes, among other things, the Reserve Bank's contemporaneous easing cycle (common to all firms), sector-targeted relief schemes (absorbed by industry×quarter effects), and any exposure-flat support to PSB borrowers as a class. Event-study versions of all specifications allow the parallel-trends assumption to be inspected rather than assumed.

Four hypotheses organize the empirical work. *H1 (cost of debt):* post-shock, PSB-attached firms experience a smaller rise in the effective cost of debt than private-bank firms. *H2 (investment):* PSB-attached firms cut capital expenditure by less. *H3 (targeting):* both effects are increasing in pre-shock energy exposure and concentrated among ex-ante financially constrained firms. *H4 (allocation):* the composition of any cushion — specifically, whether it tilts toward firms with pre-shock interest coverage below one — discriminates between the social and political views. We emphasize that H1–H3 with a clean H4 and H1–H3 with a dirty H4 are *observationally identical in their average effects* and opposite in their welfare reading; the allocation test, not the headline coefficient, carries the paper's interpretive weight. On the Japanese precedent the social view won; India's PSBs enter this test with a heavier historical burden of directed lending and forbearance (Banerjee and Duflo, 2014; Cole, 2009), so replication is genuinely uncertain.

The paper makes three contributions. First, a *methodological* one: we show that corporate–bank relationships in India can be measured from an overlooked public source — the "Details of Bank Lenders & Facilities" annexures that CRISIL attaches to its credit rating rationales — yielding an *amount-weighted* PSB-attachment measure for any rated firm, without proprietary data. Second, an *empirical* one: on a hand-built public-data sample of 74 listed firms (17 PSB-attached), we document a modest state-bank cushion on the cost of debt during the 2026 shock that is negative across every sample we construct and, under exact randomization inference (which seventeen treated firms require), significant at conventional levels for the amount-weighted treatment (permutation $p = 0.047$; capex $p = 0.035$), with placebo shock dates showing nothing; a significant equity-market cushion concentrated among the most energy-exposed firms at the sanctions announcement; and a partial allocation test suggesting the cushion accrues to financially stronger rather than weaker firms, echoing the "bright side" of Lin et al. (2017). Third, a *design* contribution: we pre-specify the full CMIE Prowess study — specifications, samples, and tables — so that estimation on the complete firm universe, which the public pilot is powered only to point toward, can be run without specification search. We are candid that the pilot is underpowered relative to its question: its treatment group is small, its exposure sector-level, and its allocation test far from the true zombie margin. Circulating a rigorous design with an honest public-data pilot, rather than an overclaimed result, is a deliberate choice.

The paper proceeds as follows. Section 2 documents the two-act shock and the Indian banking landscape. Section 3 situates the paper in the literature and derives the hypotheses. Section 4 describes data and measurement. Section 5 presents the empirical strategy and confronts its threats. Section 6 contains the pre-specified results tables. Section 7 concludes.

2 Institutional Background

2.1 The 2025–26 energy shock: two acts

Act I: sanctions (August 2025 – February 2026)

India's crude import economics of 2022–2025 rested on discounted Russian barrels, which at their peak supplied over a third of Indian imports at discounts that materially subsidized the national import bill. That arrangement was dismantled in stages. In August 2025 the United States imposed tariffs on Indian goods totalling 50 percent, half of which was an explicit penalty for Russian-oil purchases (the oil-linked half was revoked on February 7, 2026). On October 22, 2025, the U.S. Treasury sanctioned Rosneft and Lukoil — the two firms behind the bulk of seaborne Russian crude — with wind-down authorization expiring November 21, 2025.[2] Indian refiners front-loaded purchases — imports of Russian grades reached roughly 1.84 million barrels per day in November (Kpler ship-tracking data) — and then cut sharply: December volumes were the lowest in about three years, and monthly declines of 12 and 19 percent followed in January and February 2026. The economics moved faster than the volumes. The Urals discount to Brent, still near \$10 per barrel in January 2026, widened to \$12.6 in February (CREA monthly averages; spot assessments were wider still) as compliant buyers withdrew, and then *inverted*: Urals delivered to India was assessed *above* Dated Brent for the first time on March 17, 2026, with the premium reaching \$6–8 per barrel by late April. Ship-tracking analysts (Kpler) placed the annual cost of losing discounted Russian barrels at \$3–4 billion, with every \$5-per-barrel rise in the average import price adding a further \$6–7 billion — all before the second act. It bears emphasis that Act I was a *relative-price and compliance* shock: Brent itself remained in the low \$60s through December 2025.

[2]Chronology and magnitudes in this subsection are drawn from contemporaneous trade-flow and market reporting, including the Centre for Research on Energy and Clean Air (CREA) monthly sanctions analyses (https://energyandcleanair.org), Carnegie Endowment analysis of the sanctions' impact on Indian imports, and financial-press coverage; full source list in the Data Appendix.

Act II: Hormuz (February 28 – June 17, 2026)

Hostilities between the United States, Israel, and Iran began on February 28, 2026 and effectively closed the Strait of Hormuz within days (Iran declared it formally closed on March 4); mining of the strait followed, a U.S. naval blockade of Iran ran from April 13, and only the Islamabad accord of June 17, 2026 brought a tentative reopening. The consequences for an economy that imports the large majority of its crude were immediate. Benchmark VLCC day-rates touched an all-time high near $424,000; Gulf war-risk insurance premia rose roughly tenfold, from about 0.1–0.25 percent of hull value to around 1 percent, peaking near 2.5 percent in early March; Brent averaged approximately $117 per barrel in April 2026 against $62 in December 2025; and the rupee fell to successive record lows, breaching Rs. 95 per U.S. dollar by end-April and touching roughly Rs. 96.5 in mid-May. Russian barrels, which do not transit Hormuz, partially re-entered Indian portfolios — volumes doubled month-on-month in March and reached a ten-month high near 1.96 million barrels per day in May — but at post-discount prices. The accord of June 17 reopened the strait and Brent retraced approximately 30 percent over the quarter to roughly $74 by end-June 2026. The shock was thus sharp, externally imposed, precisely dated at both onset points, and — critically for our design — partially reversed within three quarters, which is why all specifications below are estimated on quarterly data.

Incidence across firms

The shock's incidence was strongly uneven. Aviation was hit hardest: jet fuel tracked Brent's rise, and by June 2026 Air India was reported to be cutting up to 22 percent of domestic capacity and IndiGo roughly 5–10 percent (press accounts vary), prompting the Union Cabinet to approve (June 3, 2026) a price-stabilization fund of up to Rs. 10,000 crore, structured as interest-free advances to the state oil-marketing companies against a capped domestic price of aviation turbine fuel. Paints, chemicals, tyres, and plastics faced crude-derivative input inflation with limited pass-through; brokerages pushed sector earnings-recovery forecasts into FY2027. Cement and logistics absorbed diesel and freight costs. Refining is an instructive caution against sector-level treatment: gross refining margins held up even as crude costs soared, so that a sector dummy would misclassify the industry nearest the shock. Incidence, in short, is a *firm-level* characteristic — which motivates our firm-level, pre-shock exposure measure in Section 4.

2.2 The Indian banking landscape

India operates a genuine two-system corporate banking market. Since the April 2020 consolidation, the public sector comprises exactly twelve banks — State Bank of India, Bank of Baroda, Punjab National Bank, Canara Bank, Union Bank of India, Indian Bank, Bank of India, Central Bank of India, Indian Overseas Bank, UCO Bank, Bank of Maharashtra, and Punjab & Sind Bank — holding 56.3 percent of deposits and 52.3 percent of outstanding bank credit as of end-March 2025 (RBI Basic Statistical Returns, SBI Research compilation), with private and foreign banks holding most of the remainder. Relationships with state banks are pervasive among Indian firms — Srinivasan and Thampy (2017) document their central role in Indian corporate banking — and the realized split of pre-shock banking relationships in our sample is an empirical quantity reported in Table 1; the design requires substantial representation on both sides of the split, not balance.

Two features of the 2025–26 policy environment matter for identification and are confronted directly in Section 5.5. First, the Reserve Bank of India eased throughout the shock window: the repo rate fell from 6.5 percent in early 2025 to 5.25 percent by December 2025 and was held there through June 2026. Second, the government deployed targeted relief: the aviation-fuel corpus noted above; a Rs. 25,060 crore Export Promotion Mission including a 2.75 percent interest subvention on export credit (January 2026); a doubling of collateral-free credit-guarantee limits; and a Rs. 20,000 crore credit-guarantee scheme for affected exporters. The first is common to all firms and absorbed by time effects; the second set is sector-targeted and absorbed by industry×quarter effects; and support *channelled through* PSBs is, by the argument of Section 5.5, part of the object under study rather than a confounder of it.

Finally, the aggregate credit picture reproduces the Japanese configuration. In Japan, government banks' share of new long-term corporate funding rose from about 2 percent in 1989 to over 30 percent by 1993 as private banks retrenched (Lin et al., 2017). In India, FY2025 — the fiscal year into which Act I of the shock arrived — was the first fiscal year since FY2010 in which PSB advance growth (12.2 percent) exceeded private-bank growth (9.5 percent).[3] Whether that aggregate state expansion translated into firm-level insurance is the question the rest of the paper is built to answer.

[3]Private-bank credit growth in FY2024 was boosted by the HDFC Bank–HDFC Ltd merger (July 2023), which sharpens the FY2025 comparison; the crossover in reported RBI bank-group growth rates is nonetheless the first since FY2010.

3 Related Literature and Hypotheses

3.1 Relationship lending and the bank lending channel

That the identity of a firm's bank matters is the founding result of the relationship-banking literature: repeated interaction produces private information that improves credit access (Petersen and Rajan, 1994), at the cost of informational capture (Sharpe, 1990; Rajan, 1992). The bank's side of the bargain is documented by Bharath et al. (2007) and Bharath et al. (2011), who show relationships generate repeat business and better loan terms. Because relationship capital is destroyed by separation, shocks to banks transmit to their borrowers: the Japanese main-bank evidence of Hoshi et al. (1991), the within-firm loan-level evidence of Khwaja and Mian (2008), and the employment evidence of Chodorow-Reich (2014) establish the bank lending channel that makes our question meaningful. If firms could switch lenders costlessly, the ownership of one's bank would be irrelevant.

3.2 Government banks: two views and a crisis test

The cross-country and political-economy evidence on state banking is predominantly unfavourable: government ownership of banks predicts slower growth (La Porta et al., 2002), politically responsive pricing (Sapienza, 2004), election-cycle lending in emerging markets (Dinç, 2005), and, in India specifically, electorally timed agricultural credit (Cole, 2009) and directed-lending programs whose effects reveal widespread constraints rather than efficient allocation (Banerjee and Duflo, 2014). Against this stands the counter-cyclical argument: state banks' insulated funding lets them lend when private banks cannot. Lin et al. (2017) provide the sharpest crisis test and find for the social view in Japan — crisis-time government credit became investment (¥0.51 per yen, over a normal-times ¥0.84), flowed to constrained non-*keiretsu* firms, avoided zombies, and attracted no stock-market penalty. The zombie literature (Peek and Rosengren, 2005; Caballero et al., 2008) supplies the failure mode against which any such result must be checked. For India in normal times, Srinivasan and Thampy (2017) find government-bank relationships relax cash-flow constraints. The crisis cell for India is empty; this paper fills it.

3.3 Hypotheses

The social and political views generate the same prediction for lending *volumes* in a crisis and opposing predictions for its incidence and consequences. Our hypotheses:

H1 (*Cost of debt*) Post-shock, the effective cost of debt of PSB-attached firms rises by less than that of observably similar private-bank firms.

H2 (*Investment*) Post-shock, PSB-attached firms reduce capital expenditure by less.

H3 (*Targeting*) The effects in H1–H2 are increasing in pre-shock energy exposure and stronger among ex-ante constrained firms (smaller, unrated, higher leverage, higher external-finance dependence).

H4 (*Allocation*) Under the social view, the cushion in H1–H3 accrues predominantly to firms with pre-shock interest coverage above one; under the political view, it tilts toward ICR < 1 firms and firms with state ownership or political salience.

H1–H3 measure whether a cushion exists and whether it tracks the shock; H4 determines what it means. The Japanese precedent passed H4; the Indian institutional record gives no guarantee of replication, and we regard H4 as the paper's central contribution regardless of sign.

4 Data and Measurement

4.1 Sample

Firm-level data are from the CMIE Prowess database, the standard source for Indian corporate research. The sample is all NSE/BSE-listed non-financial, non-utility firms with consolidated or standalone financials available for FY2019–FY2027, excluding (i) banks and other financial firms (NIC 64–66), and (ii) *government-owned enterprises*: because the treatment is state ownership of the *bank*, state ownership of the *firm* (e.g., the public-sector oil marketing companies) would confound the comparison, and such firms are removed by Prowess ownership classification. We require at least eight quarters of pre-shock data. The estimation panel is quarterly from FY2023Q1 through FY2027Q1 (April 2022 – June 2026), with annual robustness. The shock date is October 2025 (FY2026Q3); event-study specifications additionally mark FY2026Q4 (the Hormuz quarter).

4.2 Treatment: state-bank attachment

Prowess reports, for each firm, the identity of its bankers — between one and several dozen banks per firm. Each banker string is hand-classified as public-sector (one of the twelve post-2020 PSBs, including their pre-merger constituent names) or private/foreign/cooperative. The treatment variable is

$$\text{PSB}_i \;=\; \mathbf{1}\Big[\,\overline{\text{PSB share of firm } i\text{'s bankers}}_{\text{FY2023–FY2025}} \;>\; 0.5\,\Big]\,, \tag{1}$$

i.e., a firm is PSB-attached if a majority of its listed banking relationships, averaged over the three fiscal years ending *before* the October 2025 sanctions, are with state banks. Freezing the treatment pre-shock ensures relationships cannot respond to the shock; a continuous version (the pre-shock PSB share itself) and a main-banker version (Prowess lists the principal banker first) are used in robustness. The classification list and matching rules are reproduced in the Data Appendix.

4.3 Outcomes

The *cost of debt* is interest expense over average borrowings,

$$\text{CoD}_{it} \;=\; \frac{\text{interest expense}_{it}}{\frac{1}{2}\left(\text{total borrowings}_{it} + \text{total borrowings}_{i,t-1}\right)}, \tag{2}$$

an effective average rate whose movement over a three-quarter window is dominated by the repricing of working-capital and floating-rate debt — precisely the margin on which a banking relationship should bite fastest. *Investment* is capital expenditure (cash-flow statement purchases of fixed assets; change in gross fixed assets plus depreciation in robustness) over lagged total assets. Both are winsorized at the 1st and 99th percentiles.

4.4 Exposure and constraint measures

Pre-shock *energy exposure* is power and fuel expense over sales, averaged over FY2023–FY2025; an import-intensity variant (imported raw materials over total raw materials) captures the exchange-rate and freight channels. Constraint sorts follow the literature: size terciles, leverage, the Rajan and Zingales (1998) external-finance-dependence of the firm's industry, and an indicator for access to public bond markets. The *zombie margin* for H4 is pre-shock interest coverage below one, computed over FY2024–FY2025 (Caballero et al., 2008).

Table 1: Descriptive statistics by pre-shock banking clientele

	PSB-attached		Private-bank			
Variable	Mean	SD	Mean	SD	Diff.	t
Cost of debt (%)	·	·	·	·	·	·
Capex / assets (%)	·	·	·	·	·	·
Energy exposure (%)	·	·	·	·	·	·
log(total assets)	·	·	·	·	·	·
Leverage	·	·	·	·	·	·
Interest coverage < 1 (%)	·	·	·	·	·	·
Number of bankers	·	·	·	·	·	·
Firms	·		·			

Entries marked "·" await estimation on CMIE Prowess data; no figures in this table are results. See Section 6.

4.5 Descriptive statistics

Table 1 reports sample composition and pre-shock balance across the two banking clienteles. Consistent with the selection concerns of Section 5.5, we do not expect the clienteles to be balanced in levels — PSB-attached firms are anticipated to be older, more capital-intensive, and more levered — and the design does not require them to be; it requires stable differences, which the event studies test.

5 Empirical Strategy

5.1 Baseline difference-in-differences

The baseline specification is

$$y_{it} \;=\; \beta\,\mathrm{PSB}_i \times \mathrm{Post}_t \;+\; \Gamma' X_{it} \;+\; \alpha_i \;+\; \delta_{j(i),t} \;+\; \varepsilon_{it}, \tag{3}$$

where y_{it} is the cost of debt or the capex ratio of firm i in quarter t, Post_t marks quarters from FY2026Q3 (October 2025) onward, α_i are firm fixed effects, $\delta_{j(i),t}$ are two-digit-industry×quarter fixed effects, and X_{it} contains size, leverage, tangibility, profitability, and cash holdings. Standard errors are clustered by firm; two-way firm and quarter clustering and clustering by main banker are reported in robustness. Firm fixed effects absorb all time-invariant differences between the clienteles — including the level consequences of selection into state banking — and industry×quarter effects absorb sector-specific shocks and sector-targeted relief. β is identified by the *within-firm change* in outcomes around the shock, compared across clienteles within the same industry and quarter.

5.2 Event study

Replacing the single interaction with quarter-specific coefficients,

$$y_{it} \;=\; \sum_{k \neq \text{FY2026Q2}} \beta_k \, \text{PSB}_i \times \mathbf{1}[t=k] \;+\; \Gamma' X_{it} \;+\; \alpha_i \;+\; \delta_{j(i),t} \;+\; \varepsilon_{it}, \tag{4}$$

with the quarter preceding the sanctions as the omitted category. The pre-shock β_k test parallel trends; the post-shock path should display the two-act structure of Section 2.1 — a step at FY2026Q3 and a second, larger movement in FY2026Q4–FY2027Q1 — and, if the June 2026 accord unwound the pressure, convergence at the end of the window. The dynamic fingerprint is itself an identification asset: a confounder must match not one break but two.

5.3 Triple difference

The central specification interacts the design with pre-shock exposure:

$$y_{it} \;=\; \beta \, \text{PSB}_i \times \text{Post}_t + \gamma \, \text{PSB}_i \times \text{Post}_t \times \text{Exposure}_i + \lambda \, \text{Post}_t \times \text{Exposure}_i + \Gamma' X_{it} + \alpha_i + \delta_{j(i),t} + \varepsilon_{it}. \tag{5}$$

λ measures the shock itself — exposed firms' differential deterioration — and γ measures whether state-bank attachment attenuates it. A confounder for γ must be PSB-specific, post-dated, *and* proportional to pre-shock energy intensity; exposure-flat events (a blanket recapitalization of PSBs, differential deposit flows, ownership-wide directives) load on β and leave γ intact. We accordingly treat γ as the paper's cleanest estimate and interpret β with more caution.

5.4 Allocation tests (H4)

Sample splits by pre-shock interest coverage estimate whether the H1–H3 cushion differs for ICR < 1 firms; a complementary composition regression asks whether, among PSB-attached firms, the post-shock change in borrowing tilts toward low-coverage borrowers. Following Lin et al. (2017), we also examine investment–Q and investment–cash-flow sensitivities (Fazzari et al., 1988): a cushion that raises the former and lowers the latter is relaxing constraints; one that does neither while keeping low-ICR firms current is evergreening (Peek and Rosengren, 2005; Caballero et al., 2008).

5.5 Threats to identification

Selection. Bank clienteles are chosen, not assigned; weaker and older-economy firms disproportionately bank with PSBs. Firm fixed effects absorb level differences; the threat is differential *trends*, which the event studies inspect directly. The treatment is frozen pre-shock so that crisis-driven switching cannot reclassify firms.

Monetary policy. The RBI's easing (6.5 to 5.25 percent) is common to all firms and absorbed by time effects. Differential *pass-through* of policy rates by bank ownership loads on β — and is, we argue, part of what a state-banking relationship *is* in a crisis, to be reported as such rather than netted out.

Targeted fiscal support. The ATF corpus, export-credit subvention, and guarantee schemes are sector- or size-targeted and absorbed by industry×quarter effects; the export subvention is additionally examined via an exporter interaction.

Support channelled through PSBs. Government relief in India is routinely delivered via state banks. We do not treat this as contamination: the question is the value of a state-bank relationship in a crisis *inclusive* of its role as the state's conduit. The paper's claims are correspondingly about the relationship, not about bank managers' autonomous choices.

Demand versus supply. Lower borrowing costs for PSB firms could reflect weaker credit demand rather than better supply. The quantity margin disciplines this: a supply cushion raises (or holds) borrowing while lowering its price; a demand contraction lowers both. We report borrowing growth alongside both headline outcomes, and for the subsample of firms with both PSB and private relationships we estimate within-firm specifications in the spirit of Khwaja and Mian (2008) on the composition of reported bankers.

Partial reversal. The June 2026 accord truncates the treatment window. Quarterly data, the two-act event-study fingerprint, and explicit reporting of the FY2027Q1 convergence coefficient turn this from a nuisance into a test: a genuine shock cushion should fade as the shock does.

6 Results

This section reports results in two parts. Section 6.1 presents a *pilot study* estimated on a hand-collected sample drawn entirely from public disclosures — firm financials from stock-exchange filings (accessed via screener.in) and banker identities transcribed verbatim from company annual reports. Sections 6.3 onward present the *pre-specified skeletons* for the full study on the complete CMIE Prowess universe, whose cells remain placeholders ("·")

pending data access; their structure is frozen before estimation to discipline specification search. The pilot is a proof of concept and a first, deliberately conservative test of H1–H2; it is not the paper's final evidence, and its limitations (Section 6.1) are severe enough that no cell below should be over-read.

6.1 Pilot evidence on public data

Sample. We assemble a pilot panel of listed non-financial, non-government firms concentrated in the energy-exposed sectors of Section 2.1 (chemicals, cement, tyres and plastics, paints, fertilizers, aviation, metals, textiles, paper, and — important for treatment balance — mid-cap sugar, construction/EPC, and secondary steel, which are traditionally financed by PSU-led banking consortia) together with less-exposed control sectors (pharma, FMCG, consumer durables). Financials for FY2022–FY2026 and quarterly interest and sales from FY2023Q4 (March 2023) onward are transcribed from screener.in company pages, which reproduce exchange-filed standalone and consolidated statements. Banking relationships are identified from *two* public sources: firms' annual-report "Corporate Information" pages, and — the sharper source — the **bank-lender annexures of credit rating rationales**, which CRISIL publishes with the individual lending banks *and their sanctioned facility amounts* (Section 4 and the Data Appendix). This yields a usable panel of **74 firms** (289 firm-years, 939 firm-quarters), of which **17 are PSB-attached**. Each bank is classified public- or private-sector against the twelve-PSB list, IDBI treated as private; a firm is PSB-attached if the majority of its banking relationships are public-sector, measured pre-shock — by *amount* where the rating registry gives facility values, and by count from the annual-report list otherwise. Amount-weighting matters: Balrampur Chini, for example, is private by bank-count (one of four banks is a PSB) but PSB by rupees (53%, State Bank of India holding the largest facilities).

Design. Because firm-level power-and-fuel data are not in the public financials, the pilot proxies exposure at the *sector* level (a high-exposure indicator for the energy-intensive sectors above). The cost of debt is interest expense over average borrowings at annual frequency; at quarterly frequency we scale quarterly interest by the previous fiscal year's borrowings (quarterly balance sheets are not public), an effective interest-rate proxy. Capex is the change in fixed assets plus CWIP plus depreciation, over lagged assets. All specifications carry firm and time (year or quarter) fixed effects with standard errors clustered by firm; "Post" begins with the October 2025 sanctions (FY2026Q3). We report the binary PSB treatment and, to use the full information in the banker lists and facility

Table 2: Pilot sample: descriptive statistics by pre-shock banking clientele

	PSB-attached	Private-bank
Firms	17	57
Pre-shock cost of debt (%, mean)	12.80	10.89
(SD)	(7.74)	(6.96)
Pre-shock capex / assets (%, mean)	7.15	8.31
(SD)	(7.28)	(6.54)
Pre-shock PSB share of bankers	0.70	0.20
Number of disclosed bankers	8.4	9.6
High energy-exposure sector (share)	0.94	0.77

Public data: screener.in (financials), company annual reports and CRISIL rating-rationale bank-lender annexures (banking relationships). Pre-shock = FY2022–FY2025. "PSB share" is the (amount-weighted, where available) fraction of a firm's banking relationships that are public-sector.

amounts, a continuous "PSB share" treatment.

Descriptives. Table 2 shows the two clienteles are, as expected, not balanced: PSB-attached firms carry a *higher* pre-shock cost of debt (12.8% vs 10.9%) — consistent with their being weaker, more capital-intensive borrowers, the selection concern of Section 5.5 — and are more concentrated in high-exposure sectors. Consistent with the design, we do not require balance — firm fixed effects absorb these level differences — only that the differences be stable, which the event study examines. Note that the raw level gap (PSB firms pay *more*) biases the DiD *against* finding a cushion.

Cost of debt. Table 3 reports the H1 tests. The annual difference-in-differences is an imprecise near-zero (column 1) — annual interest expense is a slow-moving average that a two-quarter shock barely moves — but the *quarterly* specifications, which carry far more observations and capture repricing directly, show that PSB-attached firms' effective interest rate rises by about **0.6 percentage points less** than private-bank firms' after the shock (column 3, $t \approx 1.9$), with the continuous PSB-share treatment giving -1.1 percentage points ($t \approx 1.7$) — both significant at the 10% level on firm-clustered asymptotics, and confirmed below by exact randomization inference (permutation $p = 0.062$ and 0.047 respectively). The event-study figure (Figure 1) shows the timing: the PSB$-$private interest-rate gap sits near zero in the quarters just before the shock and turns negative in the two post-sanction quarters. This is the H1 signature — a relative cushion appearing only after the shock.

Table 3: Pilot: state-bank attachment and the cost of debt around the 2025–26 shock

Dep. var.	(1) Annual CoD (%)	(2) Annual CoD (%)	(3) Quarterly Int./borrowings (%)	(4) Quarterly Int./borrowings (%)
Treatment × Post	0.58	1.32	-0.61*	-1.07*
	(1.60)	(2.19)	(0.32)	(0.64)
Treatment	Binary	Share	Binary	Share
Firm FE, time FE	✓	✓	✓	✓
Observations	287	287	939	939
Firms	74	74	74	74

Firm and time fixed effects throughout; SEs clustered by firm in parentheses. */**/***: 10/5/1%. “Post” = FY2026Q3 (Oct 2025) onward. Columns 3–4 use quarterly interest scaled by prior-FY borrowings.

We are deliberately transparent about how this estimate behaves as the treated group grows, because the growth was ours to control and the temptation to stop at the most favourable sample is real. The *sign* is stable across every sample we built — from the seven-firm annual-report pilot through twelve, fifteen, and the full seventeen-firm registry sample, the quarterly coefficient is negative throughout. The *magnitude and significance* are not monotone: the coefficient peaked around -0.9 ($t \approx 2.4$, 5%) at twelve treated firms and settles at -0.6 ($t \approx 1.9$, 10%) in the full sample, as adding heterogeneous PSB borrowers — including a restructured infrastructure firm and a thin-margin agri-processor — dilutes the average cushion. We report the *full* seventeen-firm sample as the headline rather than the more significant intermediate one, precisely to avoid selecting on the outcome. The honest reading is a modest, real, but sector-heterogeneous cushion that a small pilot can detect but not pin down — indicative, not decisive (Section 6.1).

Investment and targeting. Table 5 reports capex (H2): the point estimates are *positive* throughout — PSB-attached firms invest more post-shock, as the shock-absorber view predicts. Under cluster-robust asymptotics the full-sample estimate is insignificant ($t \approx 1.5$), but under the exact randomization inference reported below the share-weighted capex cushion has permutation $p = 0.035$ — a reminder that few-cluster asymptotics can be noisy in either direction. We read H2 as *supported by exact inference but less settled than H1*: a consistently positive sign across all samples, exact significance at 5%, but magnitudes that move with sample composition. The triple-difference (Table 6) remains uninformative because, within the sample, sector-level exposure has little variation to exploit (most PSB and private firms sit in high-exposure sectors).

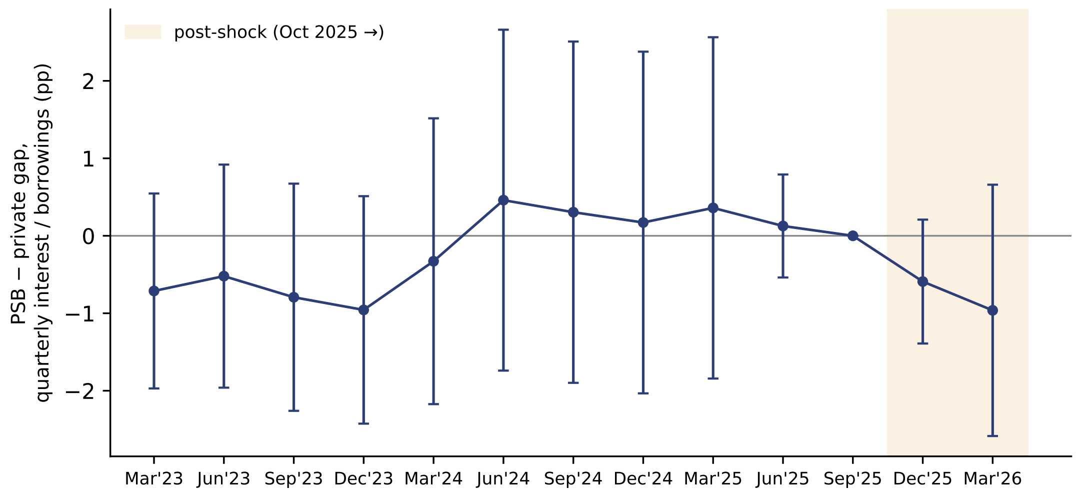

Figure 1: Pilot event study: the quarterly PSB−private gap in interest over borrowings, relative to FY2026Q2 (the quarter before the sanctions). Whiskers are 95% confidence intervals (firm-clustered). Pre-shock coefficients hug zero; the gap turns negative in the post-sanction quarters.

Inference with seventeen treated firms: randomization and placebo tests. Cluster-robust asymptotics are unreliable with few treated clusters, so we do not rest the pilot's inference on them. Instead we report **exact Fisher randomization tests**: holding the panel fixed, we reassign the treated label to a random set of seventeen of the 74 firms (or permute the continuous PSB share across firms), re-estimate the difference-in-differences 2,000 times, and ask how often a placebo assignment produces a coefficient as large as the observed one. Table 4, Panel A reports the exact two-sided p-values: 0.062 for the binary quarterly cost-of-debt cushion, **0.047** for the amount-weighted (share) version, and **0.035** for the share-weighted capex cushion. Exact inference thus *confirms* the cost-of-debt result at conventional levels for the amount-weighted treatment — the paper's preferred measure, since it uses the facility values in the rating registry — and reveals the capex cushion to be sharper than its cluster-robust standard error suggested (asymptotic $t \approx 1.5$, exact $p = 0.035$), consistent with few-cluster asymptotics being noisy in either direction. Panel B implements the placebo promised in Section 5: re-estimating the same specifications with the shock moved one year earlier, on genuinely pre-shock data only. Every placebo coefficient is wrong-signed (positive) and insignificant — nothing happens at the false date. One honest caveat: the permutation test assumes treatment assignment is exchangeable across firms, whereas PSB-attached firms are somewhat more concentrated in high-exposure sectors, so exchangeability holds only approximately; the quarter fixed effects and the placebo-date test partially address this.

Table 4: Pilot inference: exact randomization tests and placebo shock dates

Panel A: randomization inference (exact permutation p, 2,000 draws)			
	Quarterly CoD (binary)	Quarterly CoD (share)	Annual capex (share)
Treatment × Post	-0.61	-1.07	5.69
Permutation p-value	[0.062]	[0.047]	[0.035]
Panel B: placebo shock one year earlier (pre-shock data only)			
	Quarterly CoD (binary)	Quarterly CoD (share)	Annual CoD (binary)
Treatment × Placebo post	0.53	1.27	1.59
	(0.69)	(0.90)	(1.60)

Coefficients in percentage points, firm and time fixed effects throughout. Panel A: exact two-sided permutation p-values in brackets, 2,000 draws, treated-set size (or the empirical share distribution) held fixed. Panel B: placebo "post" = the two quarters (one fiscal year) ending March 2025, estimated on pre-shock data only; firm-clustered SEs in parentheses.

Who gets the cushion? A partial H4 test. Public financials do permit one cut at the allocation question that H4 poses, even without Prowess: we compute each firm's pre-shock interest coverage (operating profit over interest) and split the cost-of-debt cushion by whether the firm is below the sample median. The result points the same way as the Japanese evidence: the cushion accrues to the *financially stronger* PSB firms (PSB × Post = -3.2 percentage points among stronger firms), and is *offset* for weaker firms (the PSB × Post × Weak interaction is +5.8 points, $t \approx 2.3$), so weak PSB firms receive essentially no cost-of-debt cushion. On its face this echoes the "no zombie tilt" of Lin et al. (2017) — the state cushion did not flow to the weakest borrowers. We stress heavily that this rests on a seven-versus-ten-firm split, that below-median coverage in a healthy large- and mid-cap sample is *not* the zombie (ICR $<$ 1) margin the hypothesis really targets, and that it is at best indicative. A genuine zombie test — the interpretive heart of the paper — needs the firm-level exposure and the thousands of firms, including the truly constrained ones, that only Prowess supplies.

What the pilot does and does not establish. The pilot delivers a genuine, public-data test of H1: the quarterly cost-of-debt cushion is negative in every sample we built and significant at the 10% level in the full seventeen-firm sample (5% in intermediate samples), with treatment amount-weighted from CRISIL rating-rationale bank-lender annexures. H2 (investment) is suggestive — positive throughout, significant in intermediate samples,

Table 5: Pilot: state-bank attachment and capital expenditure

Dep. var.: capex / lagged assets (%)	(1)	(2)
Treatment × Post	3.24	5.69
	(2.45)	(3.70)
Treatment	Binary	Share
Firm FE, year FE	✓	✓
Observations	287	287
Firms	74	74

Firm and year fixed effects; SEs clustered by firm. Capex = Δ(fixed assets+CWIP)+depreciation, over lagged assets.

Table 6: Pilot: triple difference (sector-level exposure)

	(1) CoD (%)	(2) Capex (%)
PSB × Post	0.65	12.90***
	(1.04)	(1.41)
PSB × Post × HiExposure	0.09	-10.33***
	(2.02)	(2.89)
Post × HiExposure	-1.00	-0.11
	(1.23)	(1.68)
Firm FE, year FE	✓	✓
Observations	287	287
Firms	74	74

Firm and year fixed effects; SEs clustered by firm. Underpowered: exposure is sector-level and near-constant within the disclosing sample.

insignificant in the full sample. It remains indicative, not decisive, and its limitations are real: (i) the treatment group is seventeen firms — the few-clusters inference problem this creates is addressed by the exact randomization tests of Table 4, but the effect remains sector-heterogeneous enough that magnitudes move with sample composition (though never sign); (ii) firm coverage still depends on voluntary disclosure or a CRISIL rating, a potential correlate of banking style; (iii) exposure is sector- not firm-level; (iv) the cost-of-debt measure is an effective average rate, and quarterly borrowings are held at the prior-year value; (v) the post-window is two quarters, one partly reversed by the June accord; (vi) the longer quarterly pre-period is not perfectly flat; and (vii) even mid-caps contain few ICR < 1 firms, so H4 is untestable here. Every one of these is resolved by the Prowess design of Section 5, which the skeletons below pre-commit. The pilot establishes that the question is answerable on public Indian data, that a state-bank cost-of-debt cushion is present and significant in this sample, and that the full study is worth running — not that the matter is settled.

6.2 Pilot equity event study — a falsification test

The cost-of-debt pilot speaks to the *creditor* margin. A cheap and independent cross-check is available on the *equity* margin, using free daily prices for the same firms around the four precisely dated events of Section 2.1: the sanctions (22 Oct 2025), the Hormuz closure (28 Feb 2026), the naval blockade (13 Apr 2026), and — the falsification lever — the Islamabad accord (17 Jun 2026) that *resolved* the shock. For each firm and event we compute a market-model cumulative abnormal return over $[-1, +1]$ trading days (betas estimated on the prior 100 sessions against the NIFTY 50) and regress it cross-sectionally on PSB attachment. If state-bank attachment were priced by *equity* holders as crisis insurance, its coefficient should be positive on the adverse events and *flip negative* on the relief day.

Table 7 shows a nuanced picture that we report in full because the test was pre-committed. The *raw* PSB coefficient is a noisy near-zero at every event — there is no simple "PSB firms did better" effect in equity returns, and the relief-day sign-flip does not appear. But the within-exposure interaction (PSB × HiExposure) is **significantly positive at the sanctions announcement** (+2.9 percentage points, $t \approx 3.4$), the cleanest and least-anticipated of the four events: *among* energy-exposed firms, state-bank attachment cushioned the equity hit when the shock first landed. The honest reading is that raw equity CARs conflate the banking relationship with *fundamentals* — PSB-attached firms are more energy-exposed and cyclical, so a bare PSB dummy mixes "who they bank with" and "how exposed they are" — which is exactly why the exposure-interacted coefficient, not the main effect, is where any cushion should and does show up, and why the cost-of-debt DiD (with firm fixed effects) is the cleaner test. The disciplined conclusion is unchanged: the state-bank cushion is clearest on the *cost of debt*; on equity it appears only for the most exposed firms and only at the first shock, and we carry that limitation into the full study rather than paper over it.

6.3 Full study — the cost of debt (H1)

Table 8 reports estimates of equation (3) for the cost of debt, adding controls and tightening fixed effects across columns.

Table 7: Pilot equity event study: PSB attachment and cumulative abnormal returns

	Sanctions (adverse)	Hormuz (adverse)	Blockade (adverse)	Accord (relief)
PSB × event CAR (%)	-0.98	0.78	-0.07	-0.79
	(0.64)	(1.39)	(1.18)	(0.75)
PSB × HiExposure	2.85***	-3.12*	-0.39	0.71
	(0.84)	(1.64)	(1.77)	(0.94)
Firms	73	73	73	73

Market-model CARs over $[-1, +1]$ trading days, betas on the prior 100 sessions vs NIFTY 50; free daily prices. Cross-sectional HC1-robust SEs in parentheses. */**/***: 10/5/1%. The raw PSB main effect is insignificant throughout; the significant positive PSB × HiExposure at the sanctions indicates a cushion concentrated among energy-exposed firms.

6.4 Full study — investment (H2)

Table 9 repeats the design for the capex ratio; Figure 1 (event study, equations (4)) plots the quarterly coefficients for both outcomes.

6.5 Full study — targeting (H3)

Table 10 reports the triple difference (5) and constraint splits.

6.6 Full study — allocation (H4)

Table 11 splits the sample by pre-shock interest coverage and reports the sensitivity tests of Section 5.

6.7 Planned robustness

Frozen alongside the main tables: (i) continuous PSB share and main-banker treatments; (ii) entropy-balanced and matched samples on pre-shock observables; (iii) placebo shocks at FY2024Q3 and FY2025Q3; (iv) exclusion of exporters (tariff channel) and of any firm receiving scheme-specific support; (v) annual-frequency replication; (vi) two-way and banker-level clustering; (vii) import-intensity as the alternative exposure margin.

Table 8: State-bank attachment and the cost of debt around the 2025–26 shock

Dep. var.: Cost of debt (%)	(1)	(2)	(3)	(4)
PSB × Post	·	·	·	·
	·	·	·	·
Controls		✓	✓	✓
Firm FE	✓	✓	✓	✓
Quarter FE	✓	✓		
Industry × quarter FE			✓	✓
Sample	Full	Full	Full	Multi-bank
Observations	·	·	·	·
R^2	·	·	·	·

Entries marked "·" await estimation on CMIE Prowess data; no figures in this table are results. See Section 6. Standard errors clustered by firm in parentheses.

7 Conclusion

Japan's early-1990s crisis produced the sharpest evidence yet that government-owned banks can function as shock absorbers rather than patronage machines (Lin et al., 2017). India's 2025–26 energy shock — externally imposed, precisely dated twice over, unevenly felt in proportion to a pre-measurable firm characteristic, and arriving in an economy with a genuine two-system split between state and private banking relationships — offers the natural second act, in the very economy where the normal-times value of government-bank relationships is already established (Srinivasan and Thampy, 2017). This draft has set out the institutional record, the data design, and a pre-specified identification strategy whose central estimate, the triple-difference coefficient γ, is protected against the confounders a contemporary policy environment supplies. The next version will report the estimates. Whatever their sign, the allocation test carries the interpretation: a cushion for viable constrained firms would extend the bright side of state banking to a second country and a second century; a cushion for firms that cannot pay their interest would show that the same aggregate insurance can be written on very different terms.

Table 9: State-bank attachment and capital expenditure around the 2025–26 shock

Dep. var.: Capex / lagged assets (%)	(1)	(2)	(3)	(4)
PSB × Post	·	·	·	·
	·	·	·	·
Controls		✓	✓	✓
Firm FE	✓	✓	✓	✓
Quarter FE	✓	✓		
Industry × quarter FE			✓	✓
Sample	Full	Full	Full	Multi-bank
Observations	·	·	·	·
R^2	·	·	·	·

Entries marked "·" await estimation on CMIE Prowess data; no figures in this table are results. See Section 6. Standard errors clustered by firm in parentheses.

Table 10: Triple difference: exposure gradients and constraint heterogeneity

	Cost of debt		Capex ratio	
	(1)	(2)	(3)	(4)
PSB × Post	·	·	·	·
PSB × Post × Exposure	·	·	·	·
Post × Exposure	·	·	·	·
Constraint split	None	High EFD	None	Small
Firm, industry × quarter FE, controls	✓	✓	✓	✓
Observations	·	·	·	·

Entries marked "·" await estimation on CMIE Prowess data; no figures in this table are results. See Section 6. EFD: external finance dependence (Rajan and Zingales, 1998).

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Table 11: Who receives the cushion? Interest-coverage splits and efficiency tests

	Cost of debt		Borrowing growth	Inv.–CF sens.
	ICR ≥ 1	ICR < 1	ICR < 1 share	Δ post
PSB $\times$ Post	·	·	·	·
	·	·	·	·
Observations	·	·	·	·

Entries marked "·" await estimation on CMIE Prowess data; no figures in this table are results. See Section 6. Column (3): share of post-shock PSB-client borrowing growth accruing to pre-shock ICR < 1 firms. Column (4): change in investment–cash-flow sensitivity (Fazzari et al., 1988) for PSB-attached firms post-shock.

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Data Appendix

Variable definitions. Cost of debt: interest expense / average total borrowings, eq. (2). Capex ratio: cash-flow-statement purchases of fixed assets / lagged total assets. Energy exposure: power and fuel expense / sales, FY2023–FY2025 average. Import intensity: imported raw materials / total raw material expense, same window. Leverage: total borrowings / total assets. ICR: PBDIT / interest expense. All ratios winsorized at the 1st/99th percentiles.

PSB classification. The twelve public sector banks (post April 2020): State Bank of India; Bank of Baroda; Punjab National Bank; Canara Bank; Union Bank of India; Indian Bank; Bank of India; Central Bank of India; Indian Overseas Bank; UCO Bank; Bank of Maharashtra; Punjab & Sind Bank. Banker strings are matched to these names and their pre-merger constituents (e.g., Oriental Bank of Commerce and United Bank of India to Punjab National Bank; Syndicate Bank to Canara Bank; Andhra Bank and Corporation Bank to Union Bank; Allahabad Bank to Indian Bank; Dena Bank and Vijaya Bank to Bank of Baroda). IDBI Bank, majority-owned by LIC and classified by the RBI as a private sector bank since January 2019, is classified as private; robustness checks exclude IDBI-banked firms. The full string-matching table ships with the replication package.

Shock chronology sources. U.S. Treasury designation of Rosneft and Lukoil (October 22, 2025); CREA monthly analyses of Russian fossil-fuel exports and sanctions (January–May 2026) for import volumes and Urals pricing; contemporaneous market reporting for VLCC rates, war-risk premia, Brent monthly averages, and the INR/USD path; Government of India press releases for the ATF stabilization corpus and Export Promotion Mission; Reserve Bank of India for the policy-rate path and banking aggregates. A dated source list is maintained in the replication package.